

Early Warning Decision Support Component

Factor	Dataset Needed	Where to Request in Sri Lanka	Format	Resolution / Spatial Level	Purpose
Flood Hazard	Flood prediction outputs (probability, depth, extent, risk level, time)	Team Flood Prediction Model / Department of Irrigation (www.irrigation.gov.lk)	Raster / GeoJSON / CSV	GN Division / Grid level (~100m–250m)	Primary flood hazard input for joint hazard analysis and priority scoring
Landslide Hazard	Landslide prediction outputs (probability, susceptibility, risk level, time)	Team Landslide Prediction Model / National Building Research Organisation (NBRO) – (www.nbro.gov.lk)	Raster / GeoJSON / CSV	GN Division / Grid level (~100m–250m)	Primary landslide hazard input for joint hazard analysis and priority scoring
Rainfall Trigger	Rain gauge observations and historical rainfall data (intensity, duration)	Department of Meteorology (www.meteo.gov.lk)	CSV / API (Time-series)	Hourly / Daily per station	Identifies heavy rainfall events triggering floods and landslides; used for validation
Digital Elevation Model (DEM)	High-resolution elevation data (for slope, aspect, height)	Survey Department of Sri Lanka (www.survey.gov.lk) or NASA SRTM / Copernicus DEM	GeoTIFF (Raster)	10m – 30m	Derive elevation, slope, aspect; supports hazard interpretation and accessibility analysis
Population Exposure	Population counts, households, population density	Department of Census & Statistics (www.statistics.gov.lk)	CSV / Vector (Tabular)	GN Division (preferred)	Measures number of people and households exposed to hazards
Population Vulnerability	Elderly (65+), children (0–14), disabled persons, poverty / low-income indicators	Department of Census & Statistics (www.statistics.gov.lk) / Ministry of Health (www.health.gov.lk)	CSV / Vector (Tabular)	GN Division (preferred)	Calculates vulnerability index to prioritize high-risk communities
Hospital Accessibility	Hospital locations, bed capacity, ICU beds, emergency services	Ministry of Health (www.health.gov.lk)	CSV / GeoJSON (Points)	Point level (facility based)	Evaluate medical response capability and accessibility to affected areas
Evacuation Shelters	Shelter locations, maximum capacity, current occupancy (if available), facilities (water, toilets, power, etc.)	Disaster Management Centre (DMC) (www.dmc.gov.lk) / District Disaster Management Units (DDMCUs)	CSV / GeoJSON (Points)	Point level (facility based)	Identify suitable evacuation locations and available shelter capacity
Road Network & Accessibility	Road network, road classification, bridges, road condition / accessibility	Road Development Authority (RDA) (www.rda.gov.lk) / Survey Department / OpenStreetMap	Shapefile / GeoJSON (Line)	Road-segment level	Analyze accessibility for evacuation routes and emergency response
Administrative Boundaries	District, DS Division, GN Division boundaries	Survey Department of Sri Lanka (www.survey.gov.lk) / Department of Census & Statistics	Shapefile / GeoJSON (Polygon)	Administrative level	Standardizes all datasets to the same spatial reference for integration
Historical Flood & Landslide Events	Historical events (date, location, hazard type, deaths, injuries, displaced people, damages)	Disaster Management Centre (DMC) (www.dmc.gov.lk) / NBRO / EM-DAT (www.emdat.be)	CSV / GeoJSON (Polygon/Point)	District / GN Division / Event based	Validate decision support system using past disaster events and impacts
Land Use / Land Cover (Optional)	Land use / land cover classification (residential, agriculture, forest, water, industrial, etc.)	Survey Department of Sri Lanka / ESA WorldCover / Copernicus (https://worldcover2021.esa.int)	Raster / Vector	10m – 30m	Estimate exposure of different land uses and support impact assessment